Stats for Sociologists

SOC6302H1F

Fall 2026

Schedule: Thursdays, 10AM - 12PM (Lab: Fridays, 11AM - 12PM)

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Office Hours: By appointment

1 Course Description

This course introduces students to descriptive and basic inferential statistical techniques, focusing on practical applications of analyzing quantitative data through computing. Laying the groundwork for more advanced classes students may take in the future, the course covers key concepts such as summary statistics, sampling distributions, hypothesis testing, correlation, and linear regression – all framed through the lens of real-world applications. Learning is hands-on, using R statistical software to compute and interpret these techniques with publicly available data sets commonly used in social science research. Though self-paced online tutorials, interactive lectures, and lab sessions, students will gain the skills needed to analyze social science data and draw evidence-based conclusions. Students will also practice communicating statistical findings.

1.1 Course Objectives

More specifically, students who successfully complete this course should be able to…

- demonstrate introductory knowledge of statistical programming.
- select and compute descriptive and basic inferential statistics using R.
- read and interpret basic statistics as they appear in academic publications.
- communicate statistical methods and findings through reproducible reports and oral presentations.

2 Course Materials and Resources

MATERIALS

There is no required textbook for this course.

You will need to **bring a laptop computer to class**, as you are expected to interact and program throughout the lecture sessions and the lab.

SOFTWARE

The statistical programming language for this course will be R. Computations are executed from a set of typed commands which is easiest to undertake through RStudio, an editor which allows you to see your code and output. Both R and RStudio are free to download to your own personal computer:

1. Download R
2. Download RStudio (free version)

You will also need regular access to strong, reliable internet for this class.

3 Course Design

This course consists of three components: a self-paced online tutorial delivered asynchoronously, a two-hour in-person interactive lecture, and a one-hour lab.

Online Tutorial: Before most classes, there will be an online tutorial to complete at your own pace. The link to the tutorial will be available on Quercus. The asyhcnonous online tutorials will contain self-check questions (not for credit).

It is easy to get lost in a course where each week's content builds on the material of previous weeks. It is crucial to keep up with the tutorials regularly to help you stay on track.

Interactive Lecture: During the in-person interactive lecture, I will demonstrate data analysis using R and RStudio, applying the concepts of the online tutorial. Make sure to bring a laptop computer, as you are expected to interact and program along. Throughout the class, I will ask and solicit questions about the week's online tutorial, focusing on any points of confusion or tricky components.

Lab: The one-hour lab session will begin with a brief TA-led group activity focused on practice exercises drawn from the material covered in the online tutorial and lecture. The majority of the session will be self-directed, during which you are expected to work on the upcoming milestone and your individual research brief. This is a supportive, collaborative environment in which students are expected to work independently while also learning from and supporting one another. The TA will circulate throughout the lab to provide guidance, answer questions, and assist with coding and interpretation as needed. If you finish the milestone early, you are encouraged to help your peers or continue working on your research brief (attendance is mandatory).

3.1 Components

All assignments, evaluations, and tests must be submitted via Quercus.

Assessment	Worth	Due
Milestones (MS) 1-4	30%	Variable
Midterm	25%	October 21
Flash Presentation	15%	December 03
Research Brief	30%	TBD

4 Schedule

Topics and topic order are subject to changes.

Week	Dates	Tutorial	Lecture	Assignment Due
1	17-Sep	01 Introduction	R workflow	
2	24-Sep	02 Descriptive Statistics	Data Management I	
3	01-Oct	03 Sampling	Data Management II	MS01
4	08-Oct	04 Uncertainty	Statistical Tests I	
5	15-Oct	05 Associations	Statistical Tests II	MS02
6	22-Oct		**Midterm**	
7	29-Oct		Research Brief Workshop (Table 01)	
8	05-Nov	06 Presenting Data	Bivariate Regression & Figures	MS03
9	12-Nov	07 Bivariate Regression	Regression I (Table 02)	
10	19-Nov	08 OLS Multiple Regression I	Regression II	
11	26-Nov		Model Interpretation	MS04
12	03-Dec		**Flash Presentations**	
	TBD			Research Brief

5 Evaluation

5.1 Milestones (30%)

Four milestone (MS) assignments will be assigned throughout the semester. MS01 and MS04 are each worth 5%, MS02 and MS03 are each worth 10%.

Milestones will include a mixture of conceptual and applied (computational) questions. Sometimes a MS will incorporate tasks that require you to demonstrate progress on your flash presentations or research brief.

Milestones will be completed using Quarto. The submission must include the knitted pdf AND Quarto file containing the necessary code to produce the results in your MS.

Milestones are due by the beginning of class on the specified date. They should be uploaded to the designated spot on Quercus (NOT emailed to your professor or TA).

You may work together in small groups on milestone assignments. Sharing code, discussing interpretations, and helping one another understand concepts are encouraged. However, all submitted work must reflect your own understanding and effort. Writing code or answers for another student, asking someone else to write them for you, or submitting work that you do not understand or could not explain yourself is a violation of academic integrity. Completing the milestones yourself is not only the best preparation for this course (i.e. the midterm and research brief), but also for developing the skills needed to work as an independent scholar beyond the classroom.

5.2 Midterm (25%)

The midterm will consist of multiple-choice questions assessing conceptual understanding of the course material and may involve some basic calculations by hand. You will not use R during the exams; however, you will be asked to interpret R coding syntax and output.

5.3 Flash Presentation (15%)

You will give a short (approximately 5 minutes) presentation on your research brief. You will be required to submit the slides (about 5 slides) for your presentation to Quercus. Details on the flash presentation will be given at a later date.

5.4 Research Brief (30%)

You will apply the skills you've been learning to answer a research question that is of interest to you. Your Research Brief should include:

- **Question**: a clearly worded research question that can be answered with quantitative data.
- **Data**: a description of the data you are using, including a description of all variables and how they are coded. A table of descriptive statistics of the total sample and key comparison groups and a narrative of the table.
- **Plan of Analysis**: a description of your analysis (i.e., how are you going to use the data to answer the research question?)
- **Results**: a presentation of the results, including a regression table and a figure.
- **Discussion**: Provide an interpretation of the results – that is, given the results, what is the answer to the question? Are there any other possible interpretations? What limitations might there be in your data or analysis that affect how well you can answer the question?

Details on the research brief, including choice of datasets, will be given at a later date.

NOTE: To help keep you on track for your final project and receive feedback along the way, each Milestone will include a small portion related to your project–some will serve as stepping stones to guide your thinking, while others will be components that may directly contribute to your final submission.

5.5 Grading Scale

Final grades will adhere to the Arts and Sciences Graduate Grade Scale.

Research briefs will be graded as follows:

Grade	Meaning
Excellent	Strong evidence of original thinking; good organization; capacity to analyze and synthesize; superior grasp of course material with sound critical evaluations.
Good	Evidence of grasp of course material, some evidence of critical capacity and analytic ability; reasonable evidence of critical and analytic skills.
Adequate	Evidence of familiarity with course material and some evidence that critical and analytic skills have been developed.
Marginal	Some evidence of familiarity with the subject matter and weakness in critical and analytic skills.
Inadequate	Little evidence of even superficial understanding of subject matter; limited critical and analytic skills; irrelevant use of course material.

6 Course Policies

6.1 Assignment Submission

All written work must be typed, double-spaced, with 1-inch margins using 11- or 12-point Times New Roman, Aptos, Calibri, or Roboto.

Written work must be submitted via Quercus. No work will be accepted over e-mail. Files should be in PDF or .doc(x) format or as a .qmd or .R script if requested. Other file formats will not be graded and will be treated as unsubmitted assignments. It is your responsibility to ensure that the file itself is readable, which can be confirmed by downloading the file once it is uploaded to the assignment submission page on Quercus. Unreadable files are missed assignments.

6.2 Late Policy & Course Extensions

Falling behind in a statistics course generally makes learning exponentially more difficult and thus, late work is generally not accepted.

However, a due date for a Milestone assignment or the Midterm may be extended for a legitimate reason with proper documentation (e.g., illness, family emergency, religious observance, accessibility

accommodation) according to UofT's guidelines. Work will not be accepted for any assignment, or portion thereof, if the answers have already been shared with the class.

Late Research Briefs will be penalized at a rate of 2% per day up to 10% (5 days) and will not be accepted thereafter, except where there are legitimate, documented reasons beyond a student's control.

Flash Presentations, given their in class participatory nature, cannot be extended unless in extenuating circumstances in accordance with UofT's guidelines (and proper documentation). Students will be responsible for arranging and scheduling a make-up presentation session comprised of the course professor, and at a minimum two additional graduate students (classmates will also be invited but not required).

Note: Extensions beyond the term end date (e.g., the last day of a fall term class may be December 1st, but the term ends December 23rd) requires a discussion with the instructor and the graduate office, as well as completion of an SGS request for an extension of course work form. These forms will be considered by the graduate office and are not automatically approved.

6.3 Academic Integrity

Students are expected to know and adhere to the University's principles of academic integrity. Copying, plagiarizing, falsifying medical certificates, or other forms of academic misconduct will not be tolerated. Any act of plagiarism or other unethical behavior will be addressed in accordance with University guidelines. Please see the Code of Behaviour on Academic Matters for specific information on academic integrity at UofT.

6.4 Artificial Intelligence

Students are strongly discouraged from using generative AI for coding, statistical interpretation, or substantive assignment work. AI can sometimes be useful for simple technical questions, but it is often incorrect in ways that can be difficult to evaluate if you are still learning the material. A key objective of this course is to develop your ability to troubleshoot problems, interpret results, and find reliable information. Before turning to AI, make use of the online tutorials, lecture slides, and your peers and TA, all of which will often lead to a faster and deeper understanding of the material and stronger long-term skills. Students are ultimately responsible for all work they submit and are particularly encouraged to complete their own interpretation of results.

Students suspected of over reliance on AI may be required to meet with the instructor for an oral assessment to verify their understanding of the submitted work. Failure to demonstrate sufficient mastery of the material may result in no credit being awarded for the assignment.

6.5 Plagiarism Detection

Turnitin plagiarism detection software is integrated within Quercus. It uses text matching technology as a method to uphold the University's high academic integrity standards to detect any potential plagiarism. The terms that apply to the University's use of this tool are described on the Centre for Teaching Support & Innovation web site. I am not comfortable with the widespread use of this software, which effectively provides large quantities of student work to a for-profit company

whose future business uses may include using it to train machine learning and AI tools. Consequently, none of the assignments for this course are set for automatic review by Turnitin. However, if I have reason to suspect plagiarism, I may use this tool to detect plagiarism.

If you object to your work being subject to review by Turnitin, add a comment when you submit your assignment. If I do find that we need to review your assignment for plagiarism and you have asked that I not use Turnitin, I will ask that you provide sufficient secondary material (e.g., reading notes, outlines of the paper, rough drafts of the final draft, etc.) to establish that the paper you submit is truly yours. If you cannot or decline to provide the requested documentation and do not consent to us using Turnitin for review, I will decline to mark the assignment.

Plagiarism: Be careful to avoid plagiarism. It is a serious academic offense with serious penalties (see the Code of Behavior on Academic Matters). Do not present someone else's ideas as your own. Give proper references to others' ideas, and use quotation marks if you are quoting. When in doubt, err on the side of a reference. Turning in an old paper, or large parts thereof, for credit in a second (or third etc.) course, is considered an academic offense that results in students being referred off to the Office of Academic Integrity.

6.6 Accessibility and Student Accommodations

The University of Toronto is committed to accessibility. If you require accommodations or have any accessibility concerns, please visit http://studentlife.utoronto.ca/as as soon as possible.

6.7 Equity and Diversity Statement

The University of Toronto is committed to equity and respect for diversity. All members of the learning environment in this course should strive to create an atmosphere of mutual respect. As a course instructor, I will neither condone nor tolerate behaviour that undermines the dignity or self-esteem of any individual in this course and wish to be alerted to any attempt to create an intimidating or hostile environment. It is our collective responsibility to create a space that is inclusive and welcomes discussion. Discrimination, harassment and hate speech will not be tolerated. Additional information and reports on Equity and Diversity at the University of Toronto is available at http://equity.hrandequity.utoronto.ca.